

UNIT 1: OS Fundamentals

Introduction

An Operating System (OS) is system software that acts as an interface between the user and computer hardware. It manages hardware resources and provides services for application programs. This unit introduces the fundamental concepts of operating systems, their functions, and types.

Operating systems play a crucial role in ensuring efficient execution of programs, resource allocation, and system security. Understanding OS fundamentals is essential for building and managing modern computing systems.

Detailed Explanation

Functions of Operating Systems

An OS performs various functions such as process management, memory management, file handling, and device management. It ensures that resources are allocated efficiently and that multiple programs can run simultaneously.

For example, the OS schedules processes to ensure fair CPU usage.

Types of Operating Systems

Operating systems can be classified into batch systems, time-sharing systems, distributed systems, and real-time systems. Each type is designed for specific applications and requirements.

Time-sharing systems allow multiple users to interact with the system simultaneously.

System Calls and Services

System calls provide an interface for programs to interact with the OS. They enable operations such as file access, process control, and communication.

Applications

Operating systems are used in computers, smartphones, servers, and embedded systems. They are essential for running applications and managing hardware resources.

Summary

This unit introduces the basic concepts and functions of operating systems, providing a foundation for further study.

UNIT 2: Process Management

Introduction

Process management deals with the creation, scheduling, and termination of processes. A process is a program in execution, and managing processes efficiently is a key function of an OS.

This unit focuses on process concepts, scheduling, and synchronization.

Detailed Explanation

Process Concepts

A process includes code, data, and system resources. Each process has a unique identifier and states such as ready, running, and waiting.

The process control block (PCB) stores information about each process.

CPU Scheduling

Scheduling algorithms determine which process gets CPU time. Common algorithms include First Come First Serve (FCFS), Shortest Job First (SJF), and Round Robin.

Efficient scheduling improves system performance.

Process Synchronization

Synchronization ensures that processes do not interfere with each other when accessing shared resources.

Mechanisms such as semaphores and mutexes are used.

Applications

Process management is used in multitasking systems, real-time systems, and distributed computing.

Summary

This unit covers process creation, scheduling, and synchronization, which are essential for multitasking systems.

UNIT 3: Memory Management

Introduction

Memory management involves managing primary memory to ensure efficient allocation and usage. It is a critical function of an OS.

This unit focuses on memory allocation techniques and virtual memory.

Detailed Explanation

Memory Allocation Techniques

Techniques such as contiguous allocation and paging are used to manage memory.

Paging divides memory into fixed-size blocks.

Virtual Memory

Virtual memory allows programs to use more memory than physically available by using disk storage.

This improves system efficiency.

Segmentation

Segmentation divides memory into logical segments based on program structure.

Applications

Memory management is used in modern operating systems to support multitasking and efficient resource utilization.

Summary

This unit introduces memory management techniques and virtual memory concepts.

UNIT 4: File Systems

Introduction

File systems manage data storage on secondary storage devices. They provide a way to organize, store, and retrieve data efficiently.

This unit focuses on file organization and management techniques.

Detailed Explanation

File Concepts

A file is a collection of related data stored on storage devices.

File attributes include name, size, and permissions.

Directory Structure

Directories organize files into hierarchical structures.

This improves data management.

File Allocation Methods

Methods such as contiguous, linked, and indexed allocation are used.

Applications

File systems are used in operating systems, databases, and storage devices.

Summary

This unit covers file organization and management in operating systems.

UNIT 5: Deadlocks and Security

Introduction

Deadlocks occur when processes are unable to proceed due to resource conflicts. Security ensures protection of system resources and data.

This unit focuses on deadlock handling and system security.

Detailed Explanation

Deadlock Conditions

Deadlocks occur due to mutual exclusion, hold and wait, no preemption, and circular wait.

Deadlock Prevention and Avoidance

Techniques such as resource allocation graphs and Banker's algorithm are used.

Security Mechanisms

Security includes authentication, authorization, and encryption.

Applications

Deadlock handling and security are essential in operating systems and distributed systems.

Summary

This unit introduces deadlocks and security concepts for safe system operation.